# Optimal Control - Explicit MPC (Exercise)

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In this exercise we will focus on the explicit solution to the *model predictive control* (MPC) problem:

$$\begin{aligned} \underset{u(\cdot)}{\text{minimize}} \quad & \sum_{l=k}^{k+N-1} \left( x_l^T Q x_l + u_l^T R u_l \right) \\ \text{subject to} \quad & x_{l+1} = A x_l + B u_l, \\ & u_l \in \mathcal{U}, \\ & x_l \in \mathcal{X}, \end{aligned} \tag{1}$$

for which the basics are given in [1]. The MPC formulation (1) can be reformulated as a *multi-parametric quadratic program* (mpQP)

$$\begin{aligned} \underset{z}{\text{minimize}} \quad & \frac{1}{2} z^T H z \\ \text{subject to} \quad & G z \leq w + S\theta. \end{aligned} \tag{2}$$

Now, consider finding the analytical solution to the explicit MPC problem. Let the system be described by the dynamics

$$x_{k+1} = x_k + u_k, \tag{3}$$

where $\dim(x_k) = 1$ and $\dim(u_k) = 1$ .

a) Formulate the MPC problem (1) for the dynamical system (3) as an mpQP (as described in Section 8.8 [1]) for the prediction horizon $N = 2$ with the constraint $\mathcal{U} = \{u \in \mathbb{R} : |u| \leq 1\}$ on the control signal, i.e., determine $H$, $G$, $w$ and $S$ as functions of the variables $Q$ and $R$, respectively.

   *Hint:* Use some symbolic software, e.g., MAPLE or MATHEMATICA.

b) Find the solution $z^*$ to the problem derived in a) for the case when $\theta = 0$.

c) Determine the region $A\theta \leq b$ where the solution above is valid, that is, determine $A$ and $b$ (see Section 5.6 [1]), where

$$A = \begin{bmatrix} G H^{-1} G_{\mathcal{A}}^T \left( G_{\mathcal{A}} H^{-1} G_{\mathcal{A}}^T \right)^{-1} S_{\mathcal{A}} - S \\ \left( G_{\mathcal{A}} H^{-1} G_{\mathcal{A}}^T \right)^{-1} S_{\mathcal{A}} \end{bmatrix} \tag{4a}$$

$$b = \begin{bmatrix} w - G H^{-1} G_{\mathcal{A}}^T \left( G_{\mathcal{A}} H^{-1} G_{\mathcal{A}}^T \right)^{-1} w_{\mathcal{A}} \\ - \left( G_{\mathcal{A}} H^{-1} G_{\mathcal{A}}^T \right)^{-1} w_{\mathcal{A}} \end{bmatrix} \tag{4b}$$

d) Show that the optimal control is given by

$$u^* = \begin{cases} 1, & x_k \leq -\frac{Q+R}{Q} \\ -\frac{Q}{Q+R}x_k, & -\frac{Q+R}{Q} \leq x_k \leq \frac{Q+R}{Q} \\ -1, & x_k \geq \frac{Q+R}{Q} \end{cases}$$

e) Verify the solution using the MPT toolbox in MATLAB for $(Q = 2, R = 1)$ and $(Q = 1, R = 2)$.

f) We see in d) that the optimal control only depends on $R/Q$. Explain how this can be seen in a more direct way in (1).

**Solution**:

a) By introducing the vectors

$$\tilde{X} \triangleq \begin{pmatrix} x_k \\ x_{k+1} \end{pmatrix} \quad \text{and} \quad \tilde{U} \triangleq \begin{pmatrix} u_k \\ u_{k+1} \end{pmatrix},$$

the dynamic constraints in (1) may be written as

$$\tilde{X} = \underbrace{\begin{pmatrix} 1 \\ a \end{pmatrix}}_{\triangleq \Phi} x_k + \underbrace{\begin{pmatrix} 0 & 0 \\ b & 0 \end{pmatrix}}_{\triangleq \Gamma} \tilde{U}, \quad a = b = 1,$$

and the constraints on the control signal can be expressed as

$$\underbrace{\begin{pmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \\ 0 & -1 \end{pmatrix}}_{\triangleq \mathcal{E}_u} \tilde{U} \leq \underbrace{\begin{pmatrix} u_{\max} \\ -u_{\min} \\ u_{\max} \\ -u_{\min} \end{pmatrix}}_{\triangleq \mathcal{F}_u} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

Finally, introduce the weighting matrices

$$\mathcal{Q} \triangleq \begin{pmatrix} Q & 0 \\ 0 & Q \end{pmatrix} \quad \text{and} \quad \mathcal{R} \triangleq \begin{pmatrix} R & 0 \\ 0 & R \end{pmatrix}.$$

The MPC problem (1) may then be written as

$$\begin{aligned} \underset{\tilde{U}}{\text{minimize}} \quad & \tilde{U}^T H \tilde{U} + 2\tilde{U}^T F x_k + x_k^T \Phi^T \mathcal{Q} \Phi x_k \\ \text{subject to} \quad & \mathcal{E}_u \tilde{U} \leq \mathcal{F}_u, \end{aligned} \tag{5}$$

where

$$\begin{aligned} H &\triangleq \Gamma^T \mathcal{Q} \Gamma + \mathcal{R} = \begin{pmatrix} b^2 Q + R & 0 \\ 0 & R \end{pmatrix} \\ F &\triangleq \Gamma^T \mathcal{Q} \Phi = \begin{pmatrix} abQ \\ 0 \end{pmatrix}. \end{aligned}$$

Completing the squares and making the suggested change of variables [1], one ends up in the mpQP (2) with $H$ as above and

$$G \triangleq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \\ 0 & -1 \end{pmatrix}, \quad S \triangleq \frac{abQ}{b^2 Q + R} \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}, \quad w \triangleq \begin{pmatrix} u_{\max} \\ -u_{\min} \\ u_{\max} \\ -u_{\min} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

b) Since, by assumption, $0 \in [u_{\min}, u_{\max}]$, we start by solving (1) for $\theta = 0$. For $\theta = 0$, no constraints are active and $z^* = 0$. The optimal control $U$ is then given by

$$\begin{pmatrix} u_k \\ u_{k+1} \end{pmatrix} = -H^{-1} F x_k = \begin{pmatrix} -\frac{abQ}{b^2 Q + R} \\ 0 \end{pmatrix} x_k.$$

c) The region were this solution is valid is found via $A\theta \leq b$, where

$$A \triangleq \begin{pmatrix} GH^{-1}G_{\mathcal{A}}^T(G_{\mathcal{A}}H^{-1}G_{\mathcal{A}}^T)^{-1}S_{\mathcal{A}} - S \\ (G_{\mathcal{A}}H^{-1}G_{\mathcal{A}}^T)^{-1}S_{\mathcal{A}} \end{pmatrix} = \begin{pmatrix} -S \\ 0 \end{pmatrix} = \frac{abQ}{b^2Q+R}\begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix},$$

$$b \triangleq \begin{pmatrix} w + GH^{-1}G_{\mathcal{A}}^T(G_{\mathcal{A}}H^{-1}G_{\mathcal{A}}^T)^{-1}w_{\mathcal{A}} \\ (G_{\mathcal{A}}H^{-1}G_{\mathcal{A}}^T)^{-1}w_{\mathcal{A}} \end{pmatrix} = \begin{pmatrix} w \\ 0 \end{pmatrix} = \begin{pmatrix} u_{\max} \\ -u_{\min} \\ u_{\max} \\ -u_{\min} \end{pmatrix}.$$

Thus, the constraints on $\theta$ are

$$\frac{abQ}{b^2Q+R}\begin{pmatrix} -\theta \\ \theta \\ 0 \\ 0 \end{pmatrix} \leq \begin{pmatrix} u_{\max} \\ -u_{\min} \\ u_{\max} \\ -u_{\min} \end{pmatrix},$$

which are fulfilled as long as

$$-abQ\theta \leq u_{\max}(b^2Q+R), \tag{6a}$$
$$abQ\theta \leq u_{\min}(b^2Q+R). \tag{6b}$$

d) Hence, the upper and lower bounds on the polyhedral containing the origin are given by (6a) and (6b), respectively. Outside this polyhedral region the $u_k^*$ is either $u_{\min}$ or $u_{\max}$ depending on which bound (6) is not satisfied, i.e., (depending on the signs of $a$ and $b$)

$$u_k^* = \begin{cases} -\frac{abQ}{b^2Q+R}x_k, & \text{for all } \theta \text{ satisfying both bounds (6)} \\ u_{\max}, & \text{for all } \theta \text{ not satisfying (6a) (or (6b))} \\ u_{\min}, & \text{for all } \theta \text{ not satisfying (6b) (or (6a))} \end{cases} = \begin{cases} 1, & x_k \leq -\frac{Q+R}{Q} \\ -\frac{Q}{Q+R}x_k, & -\frac{Q+R}{Q} \leq x_k \leq \frac{Q+R}{Q} \\ -1, & x_k \geq \frac{Q+R}{Q} \end{cases}$$

e) The MATLAB script:

```
% Optimal Control, Homework 2, Part (e)

clear all
close all
clc

%% Define the System

model = LTISystem('A', 1, 'B', 1);
model.u.min = -1;
model.u.max = 1;

%% Define the Weighting Matrices

Q = 2;
Q = 1;
model.x.penalty = QuadFunction(Q);

R = 1;
R = 2;
```

```
model.u.penalty = QuadFunction(R);

%% Define the Prediction Horizon

N = 2;
mpc = MPCController(model, N);

%% Define the Initial Condition

x0 = -9;
x0 = 4;
%% MPT Solution

ctrl = MPCController(model, N);
loop = ClosedLoop(ctrl, model);

Nsim = 30; % Simulation Time
data = loop.simulate(x0, Nsim);
x=data.X;
u=data.U;

%% My Solution (Part d)

Myx(1)=x0;
A=1;B=1;
for k=1:Nsim

    if Myx(k)<=-(Q+R)/Q
        Myu(k)=1;
    elseif Myx(k)>=(Q+R)/Q
        Myu(k)=-1;
    else
        Myu(k)=-(Q/(Q+R))*Myx(k);
    end
    Myx(k+1)=A*Myx(k)+B*Myu(k);
end

%% Plotting the Resluts

figure(1)
subplot(211)
plot(0:Nsim,x,"Color","k","LineStyle","--","LineWidth",1.2)
hold on
plot(0:Nsim,Myx,"*","Color","k","LineWidth",1.2)
grid
xlabel('k')
ylabel('x(k)')
legend('MPT Solution','Our Solution')
title(['Closed Loop System Response using MPC with
    Q=',num2str(Q),', R=',num2str(R),',and N=',num2str(N),', and
    x_0=',num2str(x0)])

subplot(212)
plot(0:Nsim-1,u,"Color","k","LineStyle","--","LineWidth",1.2)
hold on
plot(0:Nsim-1,Myu,"*","Color","k","LineWidth",1.2)
grid
xlabel('k')
ylabel('u(k)')
```

Listing 1: MATLAB Code for Part e

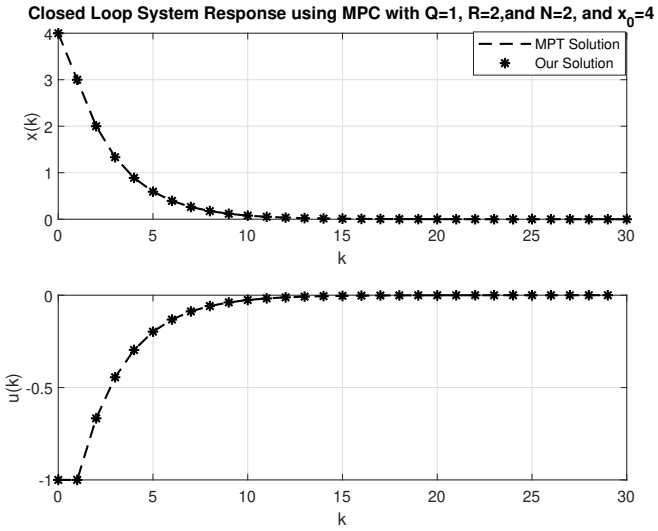



Figure 1: Comparison of the solution from MPT3 with the analytical solution

# References

[1] Hansson A. and Andersen M. Optimization for learning and control. John Wiley & Sons, 2023.